

Travel & Accommodation Booking Platform

Project Documentation Report

Frontend: React, TypeScript, Vite, Material UI
Backend: Express API deployed on Render
Author: Sali Qasarwi

1. Project Overview

Travelio is a web application for browsing hotels, searching accommodations, viewing hotel details, adding rooms to a cart, completing bookings, and managing travel reservations. The project also includes an admin dashboard for managing cities, hotels, and rooms.

The application was built to satisfy the required travel and accommodation booking platform features, including authentication, home page discovery sections, search results, hotel details, checkout and confirmation, and admin management.

2. Live Links

Resource	URL
Frontend Deployment	https://travilio-ten.vercel.app
Backend API	https://travel-and-accomodation-booking-backend.onrender.com/api
Backend Swagger Documentation	https://travel-and-accomodation-booking-backend.onrender.com/api-docs

3. Demo Accounts

Role	Username	Password	Purpose
User	<code>user</code>	<code>user</code>	Browse hotels, use cart, checkout, profile, and bookings.
Admin	<code>admin</code>	<code>admin</code>	Access admin management pages for cities, hotels, and rooms.

4. Requirements Coverage

Required Feature	Implementation in Travelio	Status
Login Page	Login form with username and password fields, validation, role based navigation, and error handling.	Implemented
Home Search Functionality	Central search bar with destination, check-in/check-out dates, adults, children, and room selection.	Implemented
Featured Deals	Displays featured hotel deals with images, hotel names, city, discount, original price, final price, and rating.	Implemented
Recently Visited Hotels	Shows recently visited hotels with image, hotel name, city, rating, and visit information.	Implemented
Trending Destinations	Displays popular destination cards using destination images and city information.	Implemented
Search Results Page	Shows matching hotels and includes filtering controls for price, star rating, amenities, and room type.	Implemented
Hotel Listings	Hotel cards include image, name, star rating, room price, city, room type, and relevant booking data.	Implemented
Hotel Gallery	Hotel details page includes a visual gallery section for hotel images.	Implemented
Hotel Information	Hotel details include name, star rating, location, description, amenities, reviews, and location section.	Implemented
Room Availability and Selection	Available rooms show room image, room type, capacity, amenities, price, availability, and add-to-cart action.	Implemented
Checkout and Payment	Checkout flow collects guest information, payment details, and special requests.	Implemented
Confirmation Page	Shows booking confirmation details and includes print/save support for the booking document.	Implemented
Admin Navigation	Admin layout includes navigation links for Cities, Hotels, and Rooms.	Implemented
Admin Search and Grids	Admin pages include search controls, detailed data grids, delete actions, and table pagination.	Implemented
Create and Update Entities	Admin create buttons and edit drawers/forms support creating and updating cities, hotels, and rooms.	Implemented

5. Main User Flows

User Flow

- User logs in using the user demo account.
- User searches for hotels by destination, dates, guests, and rooms.
- User filters and reviews available hotel results.
- User opens a hotel details page and reviews rooms, gallery, reviews, and location.
- User adds a room to the cart and continues to checkout.
- User completes guest information, payment details, and special requests.
- User receives a booking confirmation.

Admin Flow

- Admin logs in using the admin demo account.
- Admin navigates between Cities, Hotels, and Rooms.
- Admin searches and reviews entity tables.
- Admin creates new entities using form drawers.
- Admin updates existing records by selecting table rows.
- Admin deletes records using row actions.

6. Technical Stack

Area	Technology
Frontend Framework	React 19 with TypeScript
Build Tool	Vite
UI System	Material UI, MUI X Data Grid, Emotion
Routing	React Router
API Communication	Axios with a shared HTTP client
Forms and Validation	Formik and Yup
Localization	i18next and react-i18next for Arabic and English support
Testing	Vitest, React Testing Library, Playwright browser provider
Component Documentation	Storybook
Code Quality	ESLint, Prettier, Husky, lint-staged
Deployment	Vercel for frontend and Render for backend

7. Architecture and Folder Structure

The frontend uses a feature-based folder structure. This keeps each domain area close to its pages, components, API calls, types, tests, and utilities.

Key folders:

- `src/app` : application layout, providers, and router.
- `src/features/auth` : login, auth API, guards, and auth storage.
- `src/features/home` : home page sections and API calls.
- `src/features/search` : search results, filters, sorting, and search utilities.
- `src/features/hotel` : hotel details, gallery, rooms, reviews, and location.
- `src/features/cart` : cart context, storage, and cart page.
- `src/features/checkout` : checkout steps and validation.
- `src/features/confirmation` : booking confirmation display and print support.
- `src/features/admin` : admin pages, grids, forms, and management API.
- `src/shared` : reusable API client, components, theme, i18n, and utilities.

8. API Integration

The frontend reads the backend base URL from `VITE_API_URL`. The shared Axios client sends requests to the deployed Express API.

Current API base URL: `https://travel-and-accomodation-booking-backend.onrender.com/api`

Main integrated endpoint groups:

- Authentication: login/authenticate user and admin accounts.
- Home: featured deals, trending destinations, and recently visited hotels.
- Search: hotel search and amenities.
- Hotel: details, gallery, available rooms, and reviews.
- Booking: create bookings and retrieve booking details.
- Admin: navigation, cities, hotels, and rooms management.

9. UI and UX Features

The interface supports common user expectations for a booking platform:

- Clear login page with validation and API error feedback.
- Persistent top navigation for user pages.
- Admin layout with side navigation for management tasks.
- Arabic and English localization.
- Light and dark theme switching.
- Responsive layouts for desktop and mobile screens.
- Reusable cards, forms, data grids, dialogs, and toolbar patterns.

10. Testing and Quality

The project includes unit and component tests across major features. It also uses formatting and linting tools to keep the codebase consistent.

Useful commands:

- `npm run build` : production build check.
- `npm run test:run` : run the test suite once.
- `npm run lint` : run ESLint checks.
- `npm run format:check` : verify formatting.
- `npm run storybook` : view isolated component stories.

11. Deployment Summary

Frontend on Vercel

- Framework preset: Vite.
- Build command: `npm run build`.
- Output directory: `dist`.
- Environment variable: `VITE_API_URL`.

Backend on Render

- Runtime: Node.js.
- Build command: `npm install`.
- Start command: `npm start`.

- Swagger documentation available at [/api-docs](#).

12. Project Management and Tracking

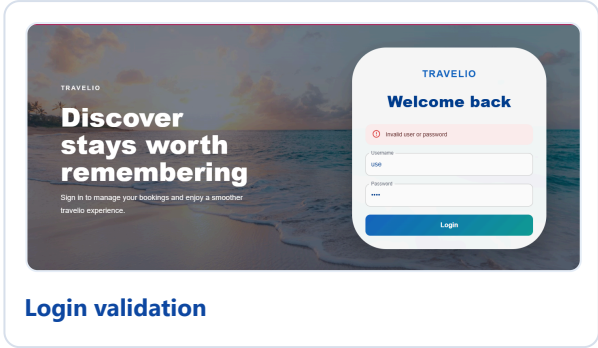
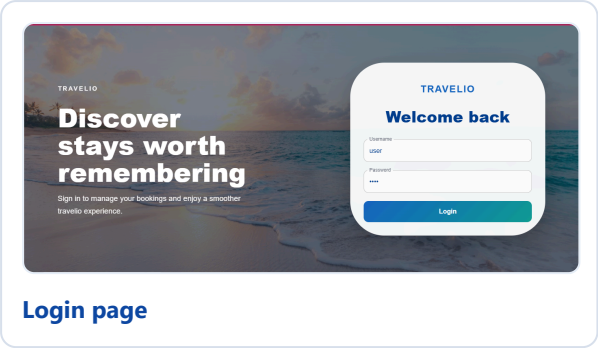
This project was planned and tracked using **Notion**. Notion was used to organize tasks, track progress, document requirements, manage feature priorities, and record implementation notes during development.

The implementation also focused on reusable components across the application, including cards, forms, data grids, dialogs, layout components, API helpers, and cart-related UI patterns.

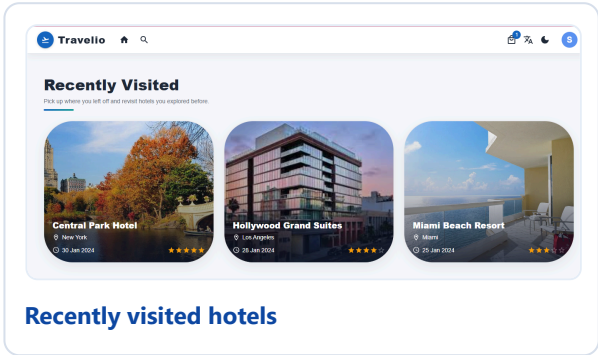
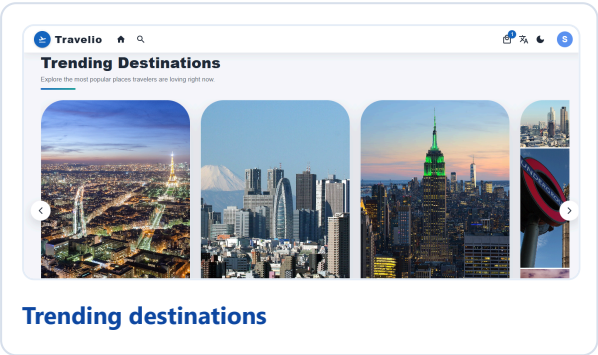
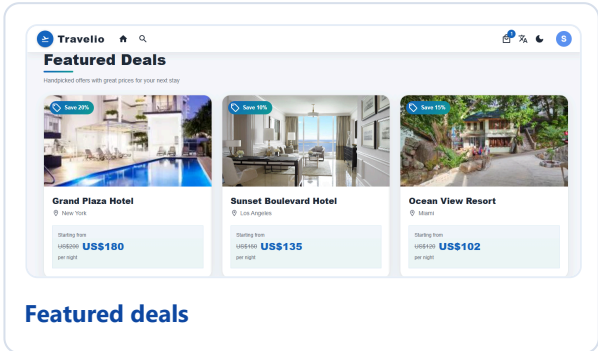
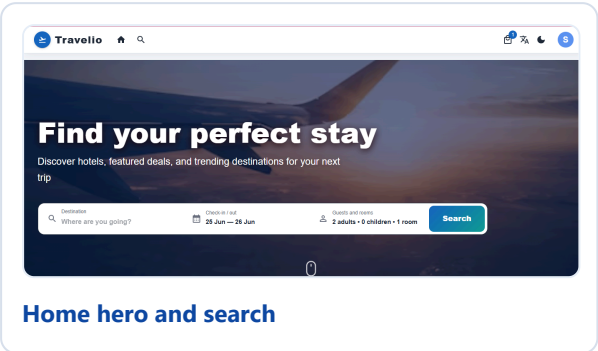
13. Screenshot Documentation

The following screenshots document the final implemented pages and states. Screenshots ending with 1 and 2 are continuation screenshots for the same page after scrolling.

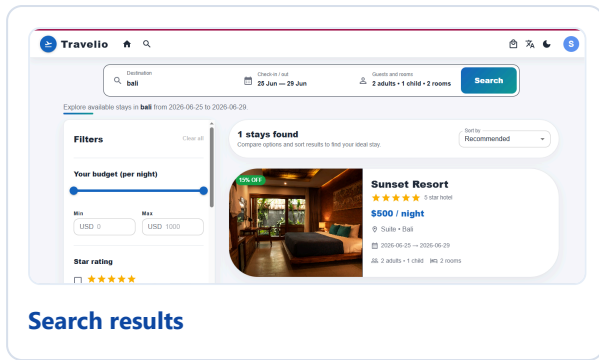
Authentication



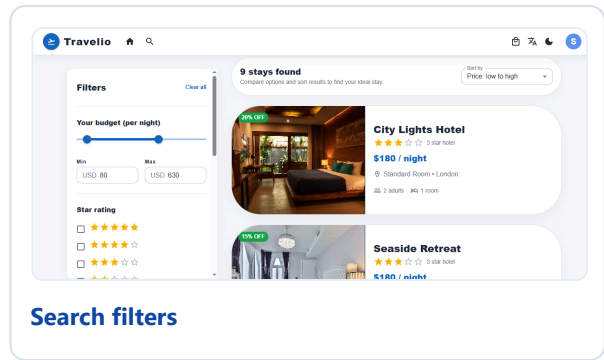
Home Page



Search

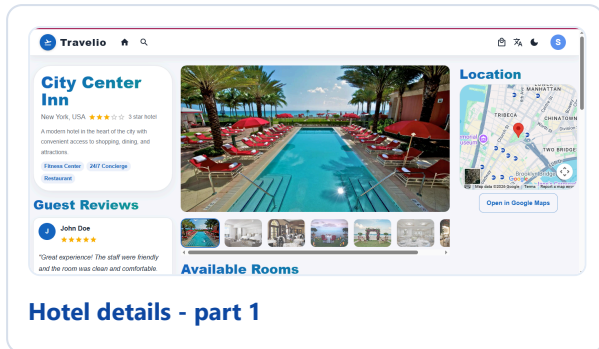


Search results

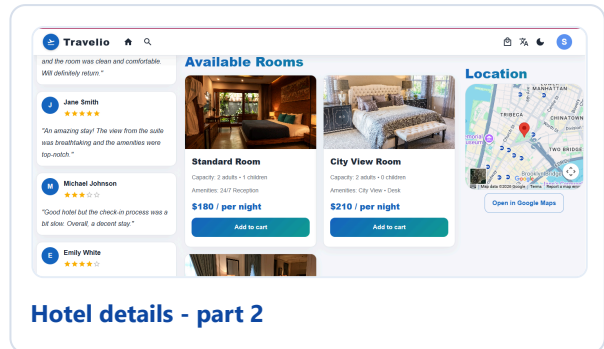


Search filters

Hotel Details

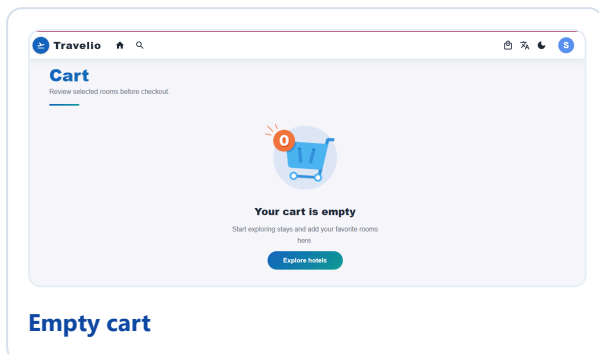


Hotel details - part 1

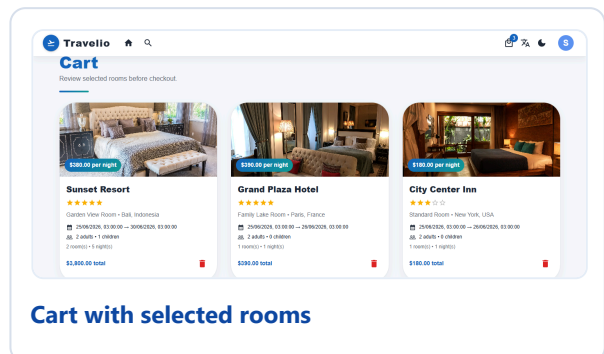


Hotel details - part 2

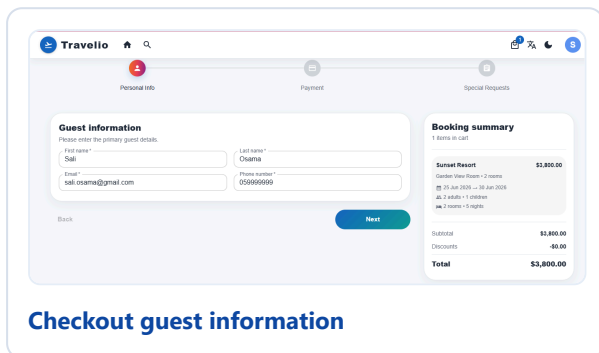
Cart and Checkout



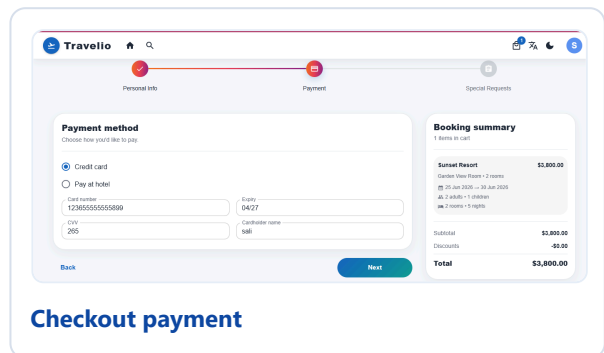
Empty cart



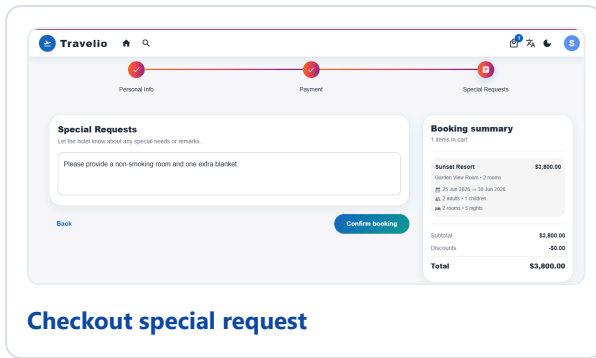
Cart with selected rooms



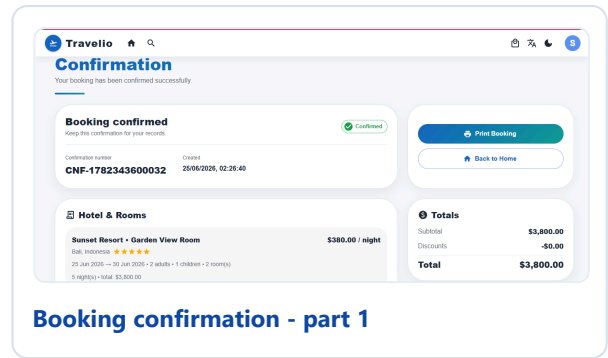
Checkout guest information



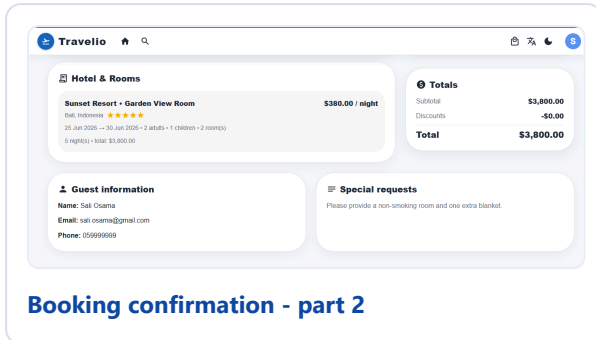
Checkout payment



Checkout special request

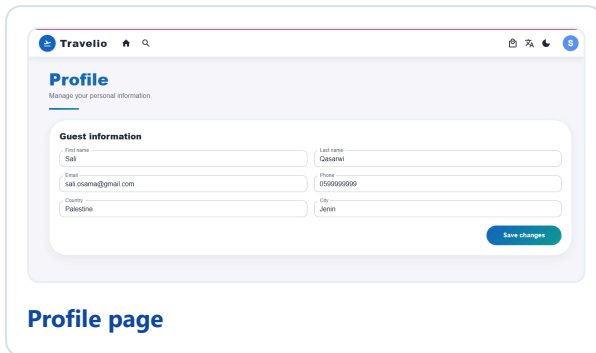


Booking confirmation - part 1

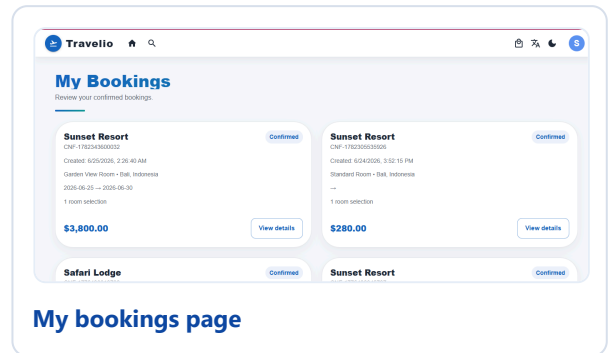


Booking confirmation - part 2

User Account

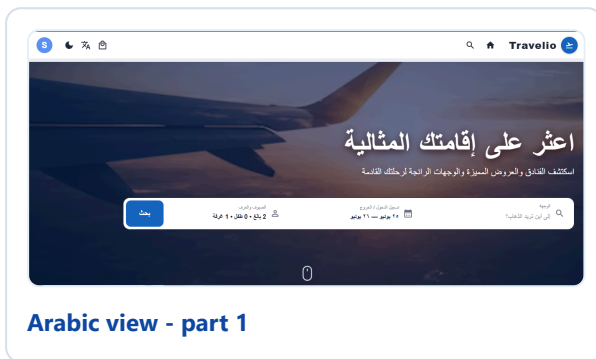


Profile page

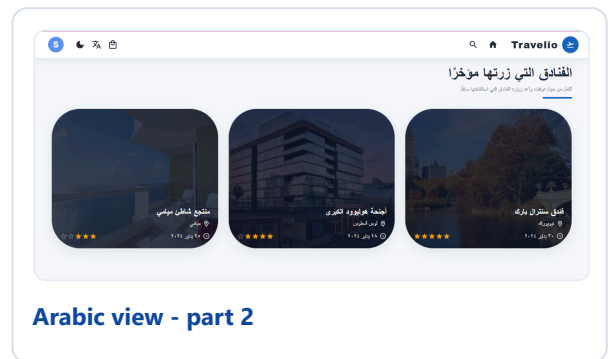


My bookings page

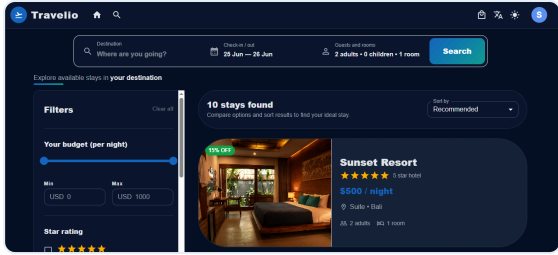
Language and Theme



Arabic view - part 1

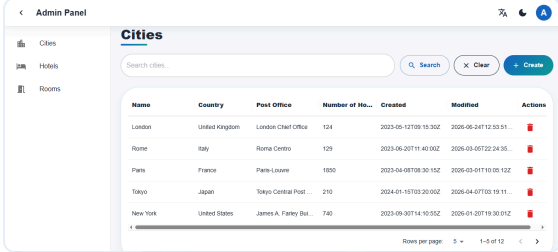


Arabic view - part 2

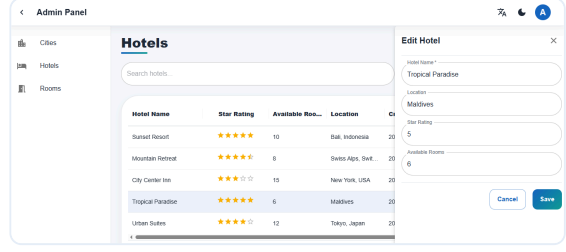


Dark theme

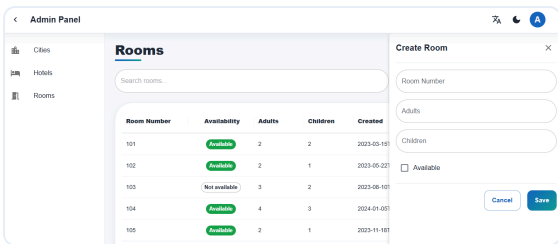
Admin Dashboard



Admin cities



Admin hotels



Admin rooms